

Applying Virtual Reality



Image courtesy of the University of Illinois

While the technology that supports virtual reality is interesting in its own right, the real payoff of virtual reality comes when it aids in solving real-world problems, provides a creative outlet, or better human existence in some way. The area in which virtual reality has been applied in order to bring about edifying personal experiences is wide and varied. These range from viewing architectural (re)creations to visualization of the human body. However, virtual reality has only begun to be applied to the myriad of possible uses.

2.1 VIRTUAL REALITY: THE MEDIUM

Virtual reality is a medium. Much like the media of music, painting, and dance, VR can be used for many purposes. A primary purpose for any medium is the communication of ideas. That is the primary focus of virtual reality in this book. The ideas can range from the purely abstract (what is it like to live inside an animated cartoon?) to the very practical (will this vehicle function appropriately after we build it?). The medium of virtual reality is much more fully explored in *Understanding Virtual Reality* [Sherman & Craig].

Although it is difficult to categorize works in any medium, it is sometimes useful to classify works for the purpose of study. This book focuses on virtual reality applications. It is helpful to classify applications to allow the reader to understand common elements, and to learn from other applications that are similar in some way to what they would like to create.

2.2 FORM AND GENRE

Form and *genre* are two terms often used to evaluate and discuss the content of media. Form is related to how the narrative is constructed and presented to the audience. An example of form from the movie *Citizen Kane* is the use of flashbacks to tell the story. Genre is a way to categorize style: Science fiction or mystery, opera or symphony, abstract or representational are all genres of particular media. Just as other media have their forms, in virtual reality there are different presentation and interaction styles that combine to make up the forms of VR. The elements of a created work are built around a basic structure that can take any number of forms (i.e., literary or interface styles). The form of an experience is manifested in the style of interface chosen for the experience; thus, in VR the form is for the most part the interface. In essence, the interface is the way things are presented, the shape of the narrative.

In VR we typically associate genre with the class of problem being addressed and form with the method of interaction and presentation. One form of interaction is the *walkthrough*. This form is a somewhat simple application interface that allows the participant to experience some model of a location through an undirected interactive narrative. Participants are allowed to move throughout the modeled virtual world. A common genre that makes use of this form is the architectural or site walkthrough, often based on a real-world location or plan, where the user can perhaps test accessibility of doors, sinks, and cabinets. However, not much else typically takes place in these spaces. One doesn't pass by the water cooler and see a conversation taking place, or hear a fire alarm, watch as people walk calmly to the nearest exit, and see them vacate the building (although one could).

In addition to the walkthrough, there are many other forms that are common VR interface forms, which we describe in the chapters that follow. Of course, the form of a VR experience is not limited to how users travel through the virtual world. Form also includes how they interact with the objects in the world. For example, to select and move an object, users might be required to make a fist near the object, move their hand to the desired location, and release their fist. An alternate form of this operation might be to point at the object with a wand, press a button, point to the new location, and release the button.

Along with these various classes of presentation/interaction styles (the forms), there is also a set of narrative styles and prototypical settings (the genres). Some virtual reality genres have begun to emerge, and, as with other aspects of VR, many are derived from other media. Examples of today's VR genres include games, scientific visualization, manufacturing procedure analysis and training, product prototyping, interactive story experience, and historical site recreations.

The choice of genre and form is orthogonal; that is, the selection of a particular genre does not (necessarily) put limitations on what form may be used. In practice, however, there may be certain combinations frequently used together; certain genres go hand in hand with specific forms. Similar forms can be used in different genres, however, as we see in the use of the newsreel and flashback in both the dramatic genre film *Citizen Kane* and the comedic genre film *Zelig*.

2.3 WHAT MAKES AN APPLICATION A GOOD CANDIDATE FOR VR

Potential new users of virtual reality should consider whether their objective will gain a sufficient advantage via this medium. Many people are attracted to virtual reality simply due to the newness of the technology. This is a perfectly valid reason to make use of virtual reality if one is tasked with investigating new technology, or if one is interested in attracting people to see your work (i.e., using VR as a means of advertising or marketing).

Apart from superficial benefits of using virtual reality, there are features of a task that can help determine whether VR can be advantageous.

Virtual reality provides a means of presenting information in a three-dimensional space, with three-dimensional input controls. Therefore, any problem that is inherently 3D can be made to work well in virtual reality. This proficiency includes any multidimensional data that can be mapped into some 3D form.

Much of what VR can do involves the creation of computer models of some real-world item or event. So, for real-world problems that rely on the use of models to achieve some goal, a virtual reality solution can often prove beneficial. Thus, model-based applications for which a computer construction can be created with fewer resources (time and/or money) than a physical mock-up would be likely to succeed as a virtual reality application. Likewise, VR is beneficial when computer models can be generated more accurately than the corresponding physical models. For example, a building can be modeled to a greater degree of detail and more lifelike scale in a computer than can be done with a hand-crafted model.

Another technological limitation that can eliminate otherwise potential applications is the degree to which the application requires input registration over a large space. Input registration refers to the ability of the hardware to maintain a stable relationship between the real world, and the sensed location of tracked objects. So, if an application requires that a user walk around a real factory, and have an augmented overlay added to their real-world view,

that application must have good registration between the real world and the augmented data to work properly. There are techniques available for attaining good registration, but the current technology generally limits this to a smaller working volume.

2.4 PROMISING APPLICATION FIELDS

Through the examination of some of the application fields that have benefited from the use of virtual reality, one can get a better sense of how well VR can be applied in other fields. “Genres” of VR applications can roughly be categorized into four broad groupings: virtual prototyping, visualization, training, and entertainment. Other genres can be classified as a subgenre of these four, such as the architectural walkthrough style of application.

2.4.1 Virtual prototyping

A classic genre of virtual reality applications that strives to save money and reduce time in bringing a product to market is *virtual prototyping*. The product can range from something small such as an appliance to large complex machinery such as a wheel loader or submarine.

A virtual prototype might be used to evaluate a product from a variety of perspectives: ergonomics, constructability, and aesthetics.

The ergonomics, or usability, of the product can be examined by presenting a virtual representation of the device to its intended users and allowing them to try it out. By virtually using the product, users can discover the little annoyances that might tip a purchase to another brand, and they can discover the big problems that could make the planned item unusable.

Some product designs are also tested for how easy it will be to construct them, and for mechanical products how easy to maintain them. By putting the product through a virtual assembly, with workers experiencing it in a realistic manner they can uncover trouble that might require a redesign of the assembly process. Likewise, a maintenance worker can be asked to perform many probable repair and inspection tasks to verify their ability to be successfully completed.

Many products get their edge in the marketplace primarily through their aesthetic qualities. Therefore, the design team assigned to consider the look of the product can get a good sense of how the product might appear to the customer by using virtual reality. Because realistic rendering is even more important to this aspect of product analysis, the experience is generally tuned for maximum visual fidelity, sacrificing some interactivity if necessary.

Not only is it easier to change a computer file than a physical structure, it is also much easier to distribute a computer file. This feature of computer-based design makes collaboration across great distances much more feasible. In a world where companies operate on a global scale, with operations and customers spread across multiple continents, the ability to quickly pass information between sites is imperative.

2.4.2 Architectural walkthroughs

Visualizing and demonstrating not-yet-constructed buildings as envisioned by their designers is a widely acknowledged, appropriate use of virtual reality. But what is it about architecture that suggests this merit? In addition to having the quality of presenting three-dimensional constructions, virtual reality improves upon traditional methods of examining a planned structure by allowing the client to experience it “first hand.” They don’t need to imagine walking into the building by looking at the model. The goal of reviewing the design prior to beginning construction (using either traditional methods or VR) is to find undesirable features in time to make changes inexpensively. In this fashion, architectural walkthroughs can be thought of as a type of virtual prototyping application.

Models can also be presented using traditional computer animation renderings, which can be used to show lighting and visual aesthetics of the space. However, by being immersed, the participant can also evaluate the scale, accessibility, and usability of the space.

A standard architectural VR application has an obvious three-dimensional representation to make the virtual world look like the real world. There is also a natural choice for navigation scheme, which is a walkthrough that allows the user to move through the space in a manner similar to how they would do so in reality. The visitor to the virtual building is constrained to walking about on the floor surfaces, but with the ability to open doors, move up and down stairs, and take the elevator.

A similar application is to create nonexistent places for reasons other than design analysis. These places might be nonexistent in time rather than space. Historical places and situations can be created as a means of allowing people to learn about the past. Here, the architectural walkthrough genre acts mostly in the realm of visualization. This exploration of data is otherwise difficult to see with unaided human senses.

2.4.3 Visualization

Computer simulations and detailed sensor inputs both can provide an overabundance of data, making it hard for researchers to perform a useful analysis. This trend previously led to the use of computer animations and then to interactive tools to aid this process. By continuing this trend to the use of virtual reality, researchers can be provided a more natural way to navigate through the space into which the data has been mapped.

Although some simulated data is not three-dimensional in nature, much of it is, so it can be easily presented via 3D VR displays. For higher-dimensional data, techniques are being developed that provide users a glimpse of the features within.

In any case, virtual reality can provide a better interface to the data than is possible through the two-dimensional interfaces of the standard desktop mouse. This interface departure can be used with nonrealistic rendering techniques afforded via computer graphics. Nonrealistic rendering is common in scientific and medical visualization, and in VR the user can be given a natural interface to view the world through a variety of different “lenses.”

2.4.4 Training

Training is also an application genre that is a rather obvious fit for VR. In fact, the oldest example of virtual reality, flight simulation, is of this genre. One basic reason for this natural fit is that mimicing the three-dimensional real world has a clear metaphor, that is, behaving in the virtual world should simulate behavior in the real world.

Virtual reality, however, allows one to do more than merely mimic reality. If reality imitation was the only goal, then it may be simpler to manufacture physical props with which a participant could practice some procedure. For example, much surgical training is performed on animals and cadavers. What virtual reality adds is the ability to practice uncommon, expensive, and dangerous tasks.

By practicing tasks in this computer-mediated system, other benefits are realized. Specifically, the simulation operator has more control over what scenarios can be presented to the trainee, and can change the scenario in response to performance. The other significant benefit is that their performance can be recorded and analyzed.

2.4.5 Entertainment

As a new medium, virtual reality has found a niche in the entertainment market. However, being new will draw people for a while, but to “keep them coming” requires an enhanced experience at an affordable price. A profitable, long-lived VR entertainment endeavor has yet to be demonstrated.

That said, the trend in computer gaming is toward more realism with techniques to render the world from the perspective of the player. This has led to an amazing improvement in the price/performance of computer graphics hardware. If this trend continues, it is likely that the cost of hardware for physically immersive displays will become low enough for many game enthusiasts to purchase them for the home.

The added feeling of immersiveness when comparing the desktop version of a game such as *Quake II* to an adaptation to the CAVE VR display indicates that the computer game community is likely to embrace virtual reality. *CAVEQuake II* is perhaps the most widely distributed of all CAVE applications.

2.4.6 Other application genres

Virtual reality has also been applied to other areas of use, such as medical, educational, and artistic uses. However, many of these uses can be categorized as subsets of the above genres.

Medical applications chiefly fall into the two categories of either visualization or training. Surgical suturing and minimally invasive surgery trainers are available commercially, and experimental work has been done on other procedures such as celiac plexus block anesthesia administration. Visualization applications include analysis of CT and MRI scan data, sometimes with the ability to aid in planning for an upcoming surgery. Work has also

been done at the University of North Carolina to allow a physician to visualize inside the patient's body while performing a procedure. Psychological treatment of phobias is a (clinician) controlled walkthrough of a visualization of some subset of the real world.

Educational applications can fall into the training, walkthrough, and visualization categories. The ScienceSpace application, a lesson in high school physics developed at the University of Houston and George Mason University, is an example of how physical processes can be visualized to allow students to come to a better grasp of the 3D nature of the subject matter than merely reading it from a book. Presently, virtual reality is being used at Valparaiso University to teach subjects such as electromagnetics and materials science. Architectural students supplement their education by creating virtual prototypes of their designs in the CAVE at the University of Illinois at Urbana-Champaign. A Japanese science museum has sought to make learning important lessons about their nation's ecology entertaining, as well as educational, by creating a virtual reality game that students participate in to learn about the ecological place of the dung beetle.

For art-oriented applications, the experience of the participant is the focus rather than the utility of what can be done in a virtual reality encounter. The goal of an artistic application may or may not be to entertain the participants, but it strives to give them a chance to view the world from another perspective and give rise to intellectual stimulation. Artistic applications also differ from many visualization, training, and virtual prototyping applications that strive to closely mimic the real world. Many artistic and entertainment applications choose to focus on the surreal or fantastic rendering and interactive possibilities offered by virtual reality.

New ideas for where to apply virtual reality are continuously arising. As virtual reality becomes more commonplace, the medium will be sought as a possible answer to many new tasks and many new artistic expressions.

2.5 DEMONSTRATED BENEFITS OF VIRTUAL REALITY

Much like any other technology, the adoption of virtual reality as a practical tool is dependent on showing that it adds some benefit unavailable through other tools. VR has promised to aid in lowering product development times, providing cost savings, scientific insight, improved learning, and engaging entertainment. In order to assess whether VR is fulfilling these hoped-for ideals, it is important to define criteria by which success is measured.

One measure is to see whether people are using the technology or not. If so, one can assume it offers something of value to the person beyond other tools they have available. If no one uses the technology, it could be because the technology is inadequate, or because it is not mature enough, still too expensive, or they haven't had an opportunity to try it.

Of course one reason to not use VR for a problem is that it is not the suitable tool for the problem. In general, VR is a tool for working with three-dimensional problems. Anytime one needs to recreate the real world (which is three-dimensional), VR is often a reasonable medium in which to work. Any task that requires the simulation of a real-world operation is a potential candidate for benefiting from VR. So, training in any number of areas is appropriate,

including medical training and military training. VR is also useful for evaluating products and places, such as new machine prototypes, architecture, and cars. Some fields in which VR has been adopted and has been used in the normal course of working include medical training applications, virtual prototyping, aircraft manufacturing (e.g., Boeing has demonstrated wearable computers with heads-up displays), and entertainment.

However, the real power of virtual reality comes from what can be done in a simulated space that cannot be done in the real world. Although VR is generally useful for imaging objects that can be represented as a 3D physical entity, like a building or a human liver, this involves representing a real-world entity exactly as it normally appears, and may not make use of all the advantages provided by VR. In VR it is possible to change the surface of the objects to show additional information such as stresses, temperature, or other attributes that are not normally visible. Similar abstracted representations can be made in aural or other senses in addition to the visual.

Usefulness can be demonstrated by organizations that have already found VR to be mature enough to be useful in its present state. Uses have included automobile and machinery design, visualization of scientific and medical data, engineering analysis, artistic expression, training, and entertainment.

In 1992, General Motors began investigating the use of virtual reality for aesthetic evaluation of automobiles as well as analysis of an automobile's ability to withstand crashes. Automobile designers began making regular use of the CAVE, escalating to the point where, by 2000, GM installed their fourth CAVE VR environment. Most other automobile manufacturers also now make use of CAVEs and similar technology and have been shown in television advertisements to illustrate that the manufacturer uses high-technology tools.

In 1994 *CRUMBS*, an application to visualize a variety of three-dimensional scientific and medical datasets, was created at the National Center for Supercomputing Applications (NCSA). Though originally developed for a specific medical application, over the subsequent years it has been expanded and used by scientists and researchers in many different fields, including oil exploration, basic physics research, genetic research, veterinary medicine, and biomedical visualization. It has also been extended in its capabilities to allow it to be a useful application in support of education by allowing an instructor to create and share scenarios with students located at a remote facility.

Nalco Fuel Tech, a company that provides engineering consulting services for operators of large boilers and incinerators, developed a prototype tool for analyzing how to improve the burn process in a way that releases fewer pollutants into the atmosphere. By visualizing the output of supercomputer simulations in a VR environment, they found that the immersive interaction of VR could aid the analysis and reduce the amount of time necessary to reach optimal design solutions. Their initial experimental effort with the Argonne National Laboratory in 1995 was rewarding enough that Nalco began their own internal VR effort in 2000.

In addition to the proven benefits of flight simulation training, both the military and NASA have found VR to be suitable for training in other areas. For NASA, training for Space Shuttle Extra-Vehicular Activity (EVA, or "space walk") missions such as the repair of the Hubble Space Telescope in 1993 and 1996, and construction of the International Space Station have

benefited from pre-flight exercises in a VR environment (beginning in 1998). One of the ways in which the military has used VR is training submarine officers in the seldom-performed task of bringing a submarine into port. A research project begun in 1994 at MIT has been successfully transitioned to a commercially supported tool for training this operation into the 21st century.

Another task that can benefit from VR training systems is surgery. Boston Dynamics, Inc. (BDI) has created a surgical task training simulator for the commonly performed task of suturing together two tubular structures (anastomosis). In addition to training for the task, an analysis mode was also created to evaluate the progress of medical students and determine when they were ready for the next phase of their training. BDI implemented the system in such a way as to make the training system match the real-world task as much as possible. The operator stood at a kiosk that mimicked standing over a patient, and held onto actual surgical instruments. The instruments were connected to a haptic feedback system that displayed forces calculated from a realistic dynamics simulation. As the operator looked down, a computer-rendered stereoscopic image hid the haptic display device, showing instead the virtual ends of the instruments interacting with the simulated tubular organs.

Photographer/artist Rita Addison turned to virtual reality as a means of expression after an automobile accident in 1992 impaired the visual centers in her brain, thus reducing her ability to use a camera. After losing her ability to express herself via the camera, in 1994 she turned to virtual reality as a means of artistic expression. In particular, working with a software developer, she produced a VR experience that gave people the ability to visually experience

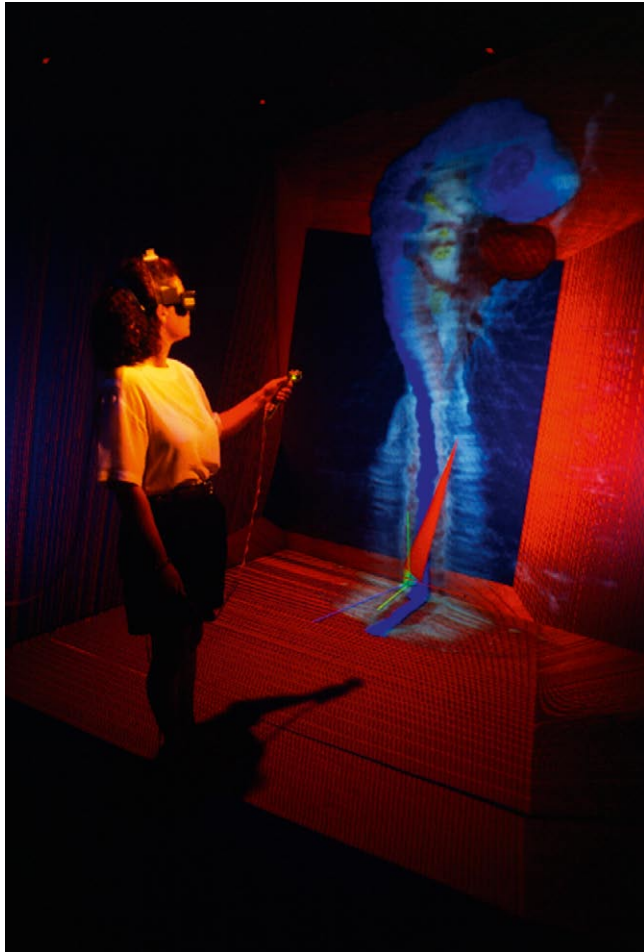


FIGURE 2-1

*A user visualizes 3D data in the CAVE using the CRUMBS application.
Image courtesy NCSA*

the world through the eyes of someone suffering the aftereffects of brain injury. Many visitors of her “Detour: Brain Deconstruction Ahead” VR experience respond emotionally, stating that they can better understand what life must be like now for Rita and other sufferers of brain injury. The experience is especially poignant for people who personally know others who have suffered a stroke or other form of brain trauma.

In 1997, Id software’s *Quake II* game was adapted for use in a CAVE environment by Paul Rajlich. Although the game was developed for play on home PCs, the increased immersion offered by the VR version demonstrates an added dimension of play. “CAVEQuake II” has been downloaded and installed on more CAVE™ and CAVE-like systems worldwide than any other application.

CAVE facilities generally do not have open public access, so for many, the DisneyQuest venue offers a more likely place to have the opportunity to experience virtual reality. The DisneyQuest family arcade centers offer a variety of interactive experiences with which families can participate. Several of the experiences consist of either head-mounted display or CAVE-like VR systems. With many of their experiences based on characters from Disney films, the creators have the opportunity to experiment with how to transition these characters from the medium of film to interactive immersive experiences.



FIGURE 2-2

Vision becomes cloudy and warped in the “Detour: Brain Deconstruction Ahead” VR art experience.
Photography by William Sherman

2.6 MORE RECENT TRENDS IN VIRTUAL REALITY APPLICATION DEVELOPMENT

Virtual reality application developers have an advantage over the pioneering scientists, artists, engineers, and educators when it comes to creating their own virtual reality experiences. Not only do they have the benefit of the experience of the early application developers, they also have a plethora of new tools that help them to do their task more quickly, effectively, and at a lower cost.

This section looks at some of the trends in application development that developers are taking advantage of.

2.6.1 Converting extant applications to virtual reality without having access to the application code

A “Holy Grail” in virtual reality application development would be to “retrofit” existing computer applications to work as virtual reality experiences without having to do any extra work, without access to the application code, and without requiring a significant level of technical expertise.

Of course, even when the technical challenges of integrating user tracking, stereoscopic display, and multisurface display are achieved, the VR developer would still need to think about the issues discussed throughout this book related to interaction schemes, navigation paradigms, and other issues related to interacting with a computer simulation in a physically immersive virtual reality environment. Fortunately, several technologies have emerged that address the issue of tackling the problems of integrating tracking, stereo, and multisurface display.

One example is a product from Mechdyne Corporation called Conduit. The goal of Conduit is to allow users of existing computer graphics applications to automatically be able to run those applications on cluster-based systems, with integrated tracking and with stereo display. The idea is that one could use an application they currently run and without access to the source code or significant code development use that same application in a CAVE, head-mounted display, or other VR device to provide immersive head tracking for the experience.

The way that this is achieved is by hijacking the application’s OpenGL calls and transforming them based on the display device and tracking information. While Mechdyne doesn’t claim to support every application that exists, they provide a list of common applications that they have tested and made “VR-aware” in their own terms.

Another tool that provides a similar functionality to Mechdyne Conduit is a suite of products from TechViz that offers a similar functionality. Though these are commercial products, it is entirely reasonable to believe that open source products with similar capabilities will be created.

2.6.2 Developing virtual reality applications with game engines

The proliferation of computer games has brought along with it a variety of game development tools. Some of these are quite useful in developing virtual reality applications. As early

**FIGURE 2-3**

Recent games like Electronic Arts' *Spore* allow the building of 3D models for creatures that even children can create. Image courtesy of Nickolaus Will

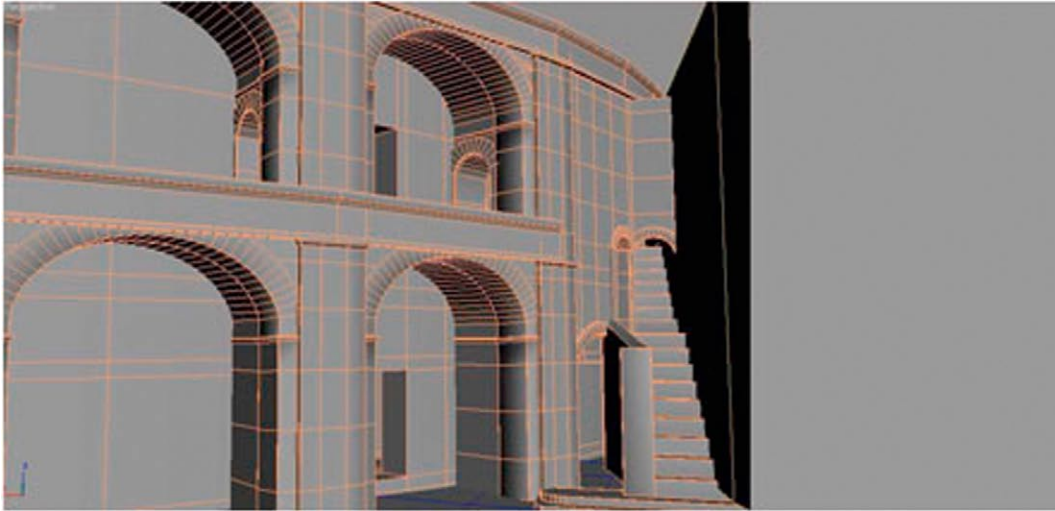
**FIGURE 2-4**

A nuclear engineering student at the University of Illinois tests a virtual model of a nuclear reactor that he created using the Unreal Editor with Unreal Tournament. Image courtesy of UIUC

as 1997, Paul Rajlich modified the computer game *Quake* to run in a CAVE. Since then, many VR applications have been developed that use game technologies.

Unreal Tournament from Epic Games and Digital Extremes is a first person shooter game in which multiple players use various weapons to engage in various battles. The game, and the Unreal Editor, provide a low-cost way for application developers to create virtual worlds for virtual reality applications. The editor is straightforward to use, and provides support for creating walk-through spaces. The game also supports animation and other useful techniques.

Professor Rizwan Uddin at the University of Illinois uses these tools to create virtual recreations of nuclear power facilities. Graduate students, undergraduate students, and even high

**FIGURE 2-5**

The Virtual Pompeii application described in Chapter 6 is being modified to work with the CaveUT software platform. Image courtesy of Jeffrey Jacobson

school students have used the Unreal Editor content creation tool to model the facilities they are interested in.

CaveUT is an adaptation of the *Unreal Tournament 2004* game engine that works with multiwall virtual reality display environments, including tracking of the users' point of view and also supports dome-style displays. The Virtual Pompeii application described earlier is being converted to work with the Unreal game engine and will be compatible with CaveUT.

2.6.3 Low-cost input devices

Another positive outcome of the popularity of video games is that the gaming industry has developed a variety of low-cost, durable game controllers. One that is particularly useful for virtual reality applications is the Nintendo Wii Remote (otherwise known as the Wiimote).

The Wiimote is a handheld device that an end-user can use to provide input to games that run on the Wii console. VR developers

**FIGURE 2-6**

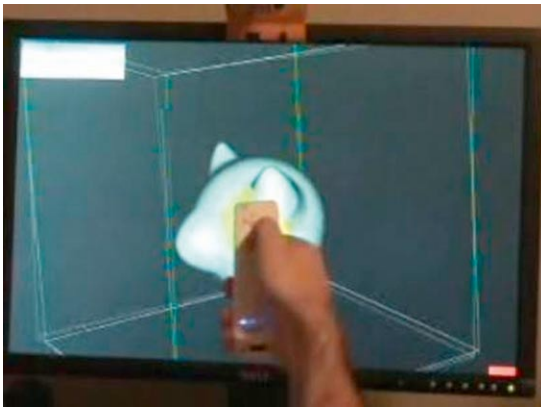
CaveUT supports multiwall projection environments.

**FIGURE 2-7**

The Wii Remote has caused considerable change to the nature of input devices in commodity console game systems by providing the types of kinesthetic inputs VR developers have used for decades.

quickly recognized its potential as a VR input device and started creating applications that support the Wiimote.

The Wiimote provides sensing of acceleration on three axes, as well as an optical sensor that is used in conjunction with a stationary Sensor Bar to provide information on the direction the Wiimote is pointing. In addition to its utility as an input device, the Wiimote has a built-in vibrator that the application developer can control. The low cost, combined with the functionality that the device provides, as well as the fact that the Wiimote operates without being encumbered by a wire makes the Wiimote an attractive option for many applications.

**FIGURE 2-8**

A Wiimote being used as an input device to a fishtank VR system.

2.6.4 Cluster-based compute engines and high-performance graphics acceleration

Many of the applications described in this book relied on high-cost graphics supercomputers. Application developers today have the advantage of the vastly improved price/performance ratio that has resulted from the mass marketing of computing and graphics technology.

Modern graphics cards, with very reasonable prices, provide an amazing level of performance compared to the technology of yesteryear that cost orders of magnitude

more. The computer game industry has pushed the capabilities of the graphics cards even at price points that are attractive to the average mass-market consumer.

Likewise, the price of computers has decreased to such a level that the typical home computer now possesses more computational capability than the supercomputers of decades past.

Many virtual reality developers are taking advantage of the idea of clustering several low-cost computers together to gain computational capability. One scenario that is appearing more and more frequently is that multiwall projection environments, like CAVEs, are being driven by a computational cluster that has one computer and one high-performance graphics card per display surface. Other scenarios include using a multiprocessor computer with several graphics cards, using a handful of multiple processor computers, and others.

The trend toward using a computer per display wall is somewhat ironic in that the original CAVE environments were driven in that exact way, using a computer per wall, where the computers were interconnected with SCRAMNet (Shared Common Random Access Memory Network). As graphics supercomputers became available, the model was to use a graphics supercomputer with multiple graphics “pipes” or hardware accelerators.

Many virtual reality software libraries now provide support for using clusters of computers rather than a monolithic computation engine. Additionally, software solutions such as the Mechdyne Conduit provide a way to utilize extant software applications on clusters of computers with little or no modification.

2.6.5 Passive stereo displays

In the past, many projection-based VR environments used active stereo glasses, and for good reason. Active stereo works by time multiplexing the left eye view with the right eye view. The glasses shutter in synchrony with the display of each eye’s view. A major advantage of active stereo glasses is that they work even when there is not an inherent “up” or “down” of the scene. That is, no matter which way a participant positions their head with respect to the display, the stereo still functions correctly. For example, the floor in a CAVE has no inherent up or down. Consequently, the shuttering glasses work nicely because they function no matter how they are oriented with respect to the display. A major disadvantage of active, time-multiplexed stereo is cost.



FIGURE 2-9

A small-scale computing cluster that drives the CUBE (six-sided CAVE-like device) at the Beckman Institute using the Syzygy virtual reality library.

**FIGURE 2-10**

A large scale computing cluster.

In an effort to reduce costs, many virtual reality environments now use passive stereo techniques. The most common passive stereo technique is using polarizing filters on the projection source, and corresponding filters in the participant's glasses. The most common polarization scheme is where the images intended for one eye are polarized horizontally, and the images intended for the other eye are polarized vertically. This works nicely as long as the display has an inherent up and down, and the participant does not tilt their head. If the participant tilts their head, the result is unsatisfactory.

One way to overcome the problems of linear polarizing filters is to use circularly polarized filters (polarized clockwise and counterclockwise). They have the advantage of not being affected by head tilt, but suffer from a worse rejection ratio.

Another passive stereo technique is to use anaglyphic stereo. The red and blue filters that were used in old-fashioned three-dimensional movies are an example of this technique. That technique, though, suffers from the problems inherent with filtering two colors. A more modern spin on the red/blue filters is implemented by a technique known as Infitec™ stereo separation. In this technique, filters are created that allow a wide range of colors to pass, but with a selected subset of each color wavelength being filtered. The filter for the other eye also allows the wide range of colors, but with a different subset of wavelengths being filtered. The end result is that users perceive the full color spectrum, but the left eye receives the image intended for it, and the right eye does as well. This technique does not suffer problems with head tilt.



FIGURE 2-11

This example of handheld augmented reality from the Christian Doppler Laboratory at the Graz University of Technology allows the participants to use their handheld PDA devices to “see” a virtual train on the physical toy train track. Images courtesy of Vienna University of Technology

2.6.6 Augmented reality and handheld devices

Although augmented reality (AR) is not a new idea, new applications are being developed at a rapid pace. One area that is relatively new is using cellular telephones and/or other handheld displays as the interaction device. The Wellington Zoo Advertising Application is an example of hand held augmented reality that uses a cellular telephone as the viewing device. The telephone also has a camera built into it, so it serves as a fully functioning AR system.

2.6.7 Wireless interaction and optical tracking

One limiting factor in many of the foundational applications described in this book is that the participant is encumbered with devices and wires. Technology can solve that problem, and as technology progresses we expect to see many more examples of unencumbered virtual reality.

One area that has made tremendous strides in freeing the participant from wires is in the area of tracking. Optical tracking, which uses cameras to “watch” the tracked object or person, is becoming quite useful as a means of tracking without wires. In order to be most useful, optical tracking must utilize more than one camera in order to determine depth and to solve the problem of occlusion between objects (when something else comes between the tracked object and the camera, making it impossible for the camera to see the object).

**FIGURE 2-12**

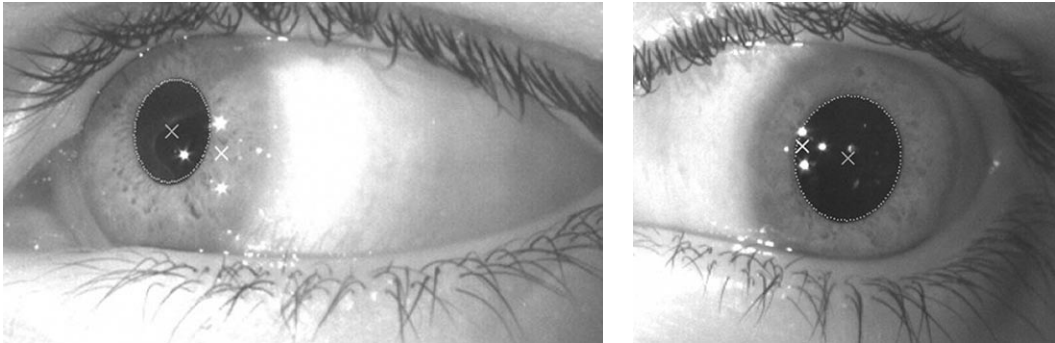
This person is using their cellular telephone to “see” an augmented reality soccer game. The AR application is allowing the user to interact with the virtual ball with their physical foot.

**FIGURE 2-13**

Current optical tracking systems determine the coordinates of the tracked entity. A camera “watches” one or more identifying marks in the scene. In order to aid the system, reflective markers are often used. The two styrofoam balls mounted on the glasses are covered with retroreflective tape so they appear as the brightest objects in the scene. The camera(s) can determine information about this person’s location and orientation by where the markers appear in the camera’s view. Image courtesy of Valparaiso University Visualization Laboratory

Currently, many optical tracking systems use retroreflective tape or markers to make it easier for the camera to identify the tracked object. As research advances, optical tracking will be useful without requiring any markers.

Another advance is in the area of eye-tracking methods. By tracking the eyes, computational resources can be deployed where the participant is looking.

**FIGURE 2-14**

Eye tracking allows the application developer to utilize information about where the participant is looking, as opposed to just the direction their face is pointing.

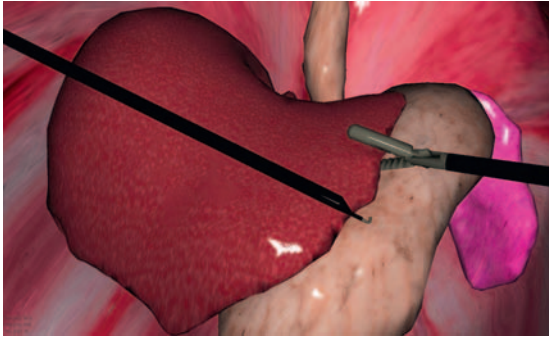
**FIGURE 2-15**

Although still not common, some application developers are including treadmills into their applications for a more natural running/walking sensation. The image in the left shows a manifestation of an omnidirectional treadmill built in the form of a large hamster exercise ball. The image on the right is implemented as a treadmill of treadmills with additional technology.

Other devices can be made wireless through many different wireless technologies such as radio, ultrasound, or infrared communication.

2.6.8 More senses

Though it is still the case that most virtual reality applications rely primarily on visual displays, and to a lesser degree to aural displays, more sensory displays are becoming more common. Haptics is an area that is becoming more widely utilized.

**FIGURE 2-16**

Haptics is becoming more widely used in VR applications. In this example, surgical tools provide haptic feedback in similar ways to the BDI Surgical Simulator described in Chapter 5.

Image courtesy of Anderson Maciel (INF/UFRGS) and Suvranu De (RPI).

**FIGURE 2-17**

Multiperson augmented reality games can be played outdoors. Of course, people who are not in the game will not see the objects that are augmenting the view of the players.

2.6.9 Multiperson virtual and augmented reality

A major trend in computer gaming that has also followed in virtual reality and augmented reality is the inclusion of multiple participants. It is only natural to want to share your experience with other people, and it is often highly useful to collaborate with others in a virtual reality environment. By building VR applications with computer game engines, multiple people can participate in the virtual experience. While many of the multiperson virtual reality experiences were built without the use of game engines, for those that use the game engines, the multiperson characteristics come with little or no extra effort.

2.6.10 Tiled displays

One limitation of older virtual reality technology is that it suffers from poor resolution and is limited in brightness. Modern head-mounted displays are an improvement, as are modern projectors. There is a trend, though, to achieve higher brightness and more pixels by tiling multiple display systems together. This can be achieved by projecting multiple projectors onto a single display surface, or by abutting multiple displays together.

Of course, achieving a good alignment and registration, as well as color and brightness matching, is still a nontrivial engineering challenge.

Tiling together multiple flatscreen displays can be problematic if the bezels and frames are obtrusive for your application. Tiling projectors together to project on a single screen (typically, rear projected) brings with it the difficulty in aligning the displays. Image alignment can be carried out



FIGURE 2-18

This display was created by tiling together multiple flatscreen display monitors. The net effect is high resolution, and plenty of screen real estate.

Courtesy of Visbox, Inc.

mechanically (to line up the edges physically) or in software by allowing the images to overlap and using software to do edge blending.

Of course, the goal of tiling displays together is to achieve high pixel density with appropriate screen real estate. As display technology improves, there will be less incentive to tile multiple displays together.

2.6.11 Head-based projection

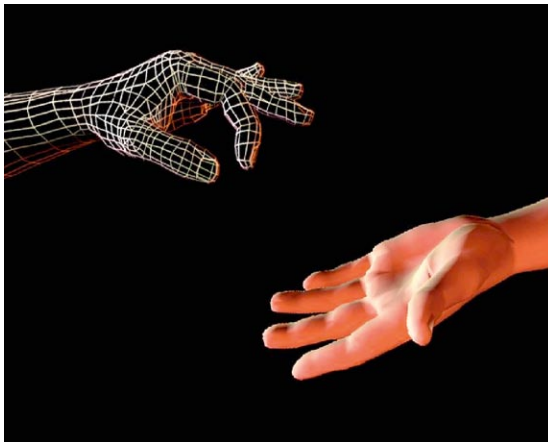
With head-based projection, the participant wears a small projector on their body or head. The net result is that there is a display wherever the participant looks. One example application would be to integrate head-based projection with augmented reality technology. A camera could “recognize” the person you are talking to, and project that person’s name on their body to give you a reminder of their name.

2.6.12 Tele-immersion

Tele-immersion allows multiple people to interact together as though they are in the same place. Instead of using avatars to represent the participants, the system creates three-dimensional models of each participant in real time and reconstructs them at the other end of the system.

**FIGURE 2-19**

This picture shows a tiled display that is rear projected. The projectors are mechanically aligned. Image courtesy of Visbox, Inc.

**FIGURE 2-20**

Tele-immersion allows users who are not located in the same place to interact as though they are. The system uses computer vision to interpret the scene, which is then integrated with the other locations and then reconstructed as a single environment.

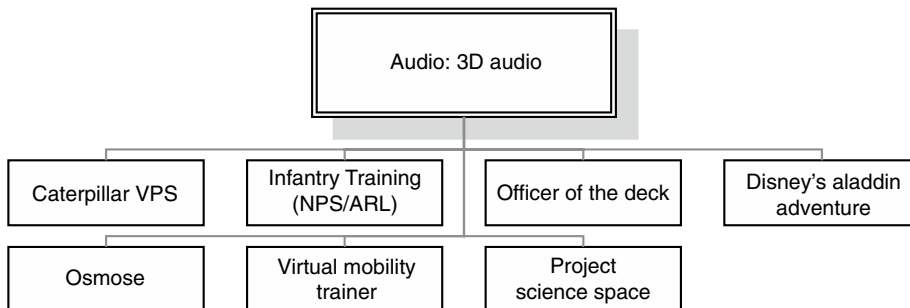
2.7 A FRAMEWORK FOR VR APPLICATION DEVELOPMENT

This book presents several exemplary VR applications that are either inspirational or foundational in nature. The casual reader may be interested in a specific genre, or may have a specific application to develop for a narrow focus. Regardless, lessons learned in one area often can impact a topic in another area. To help in making sense of the following applications, we have developed a taxonomy and database to cross-reference applications, or to do searches for specific information that you may be interested in. This is described in detail in the final chapter. The database is available for access on the companion website.

**FIGURE 2-21**

A game player immerses himself into the game world.

Photography by William Sherman

**FIGURE 2-22**

Results of searching for 3D audio show that a wide range of applications in different areas use the technology.

An example of how this can be useful is finding commonalities between applications in different fields. For example, you may be interested in 3D audio for entertainment VR applications. However, applications in business, military, science, and art all use 3D audio. By searching on such a topic in the database, you would find that the following applications all use 3D audio.

We have defined the fields for the database based on high-level categories. For each of the applications, their characteristics are identified. Again, we refer the reader to the final chapter for a full treatment of this topic. The main fields and their subcategories are:

- **Audio:** 3D audio, Non-speech, Sonification, World Referenced, Head Referenced
- **Direct Selection:** Pointer, Gaze, Crosshair, Torso, Device, Coordinate, Landmark
- **Display:** Front Projection, Rear Projection, Mono, Stereo, Reach-in, Computer Monitor
- **Genre:** Site Exploration, Game, Virtual Prototyping, Visualization, Training
- **Goals:** Analysis of 3D Data, Cost Savings, Profit, Improve Quality of Life, Artistic Expression, Informative Expression, Entertainment/Escapism, Simulation Experimentation, Safety, Marketing, Training, Education, Virtual World Creation, Modify Objects
- **Head-based:** HMD, Opaque, Video See-through, Optical See-through, BOOM, PUSH
- **Input:** Head-tracking, Prop-in-hand Tracking, Hand-tracking, Two Limbs Tracked, Two Hands Tracked, Feet Tracked, Objects/Props Tracked, 6DOF, 3DOF, Other DOF, Electromagnetic, Mechanical, Optical, Videometric, Ultrasonic, Inertial, Neural, Fingers, Eyes, Predictive Analysis
- **Item Selection:** Contact, Contact by Intermediary, Point, 3D Cursor, Aperture, Menu, MiniWorld, Name
- **Manipulation:** Direct User Control, Physical Control, Virtual Control, Agent Control, Machine Control, Ratcheting, Artificial Constraints, Natural Constraints, Distance AAAD, Local Action-Direct Action, Local Action-Remote Effect, Action through an intermediary, Laser Pointer Beam Scope, Expanding Pointer Beam, Adjustable Pointer Beam, Hysteresis, EGO Reference Frame, EXO Reference Frame, Feedback,
- **Navigation:** Path Following, Maps, Landmarks, Placenames, Bread Crumbs, Compass, Instruments, Exocentric View, Coordinate Display and Grid, Constrained Travel
- **Objects:** Geography, Agents, Artifacts, UI Objects
- **Physical Input:** Correction Table, Buttons, Switches, Valuator, Wand, Joystick, 6DOF, Props
- **Physics:** Static World, Cartoon Physics, Newtonian, Aristotelian, Choreographed, Collision Detection, World Persistence, Dynamic World, Inference from Real World, Computational Model
- **Platform:** Ring, Kiosk, Ambulatory, Cockpit
- **Point-of-View:** First Person, Second Person, Third Person, Inside-out, Outside-in
- **Representation:** Realistic, Abstract, Cartoon, Mixture, Qualitative, Quantitative
- **Sharing:** One immersed with onlookers, Two or more immersed, Open Display, Multiperson Cockpit, Collaboration, Avatar Representations, Synchronous

Collaboration, Asynchronous Collaboration, Annotation, World Congruity
Synchronous, World Congruity Asynchronous, Doctor/Patient

- **Speech Recognition:** Speaker Independent, Speaker Dependent, Push-to-talk, Name-to-talk, Look-to-talk
- **Stereo Display:** Time Interlaced, Passive, Linearly Polarized, Circularly Polarized, Anaglyphic, Spatially Multiplexed
- **Travel:** Physical Locomotion, Redirected Walking, Ride Along, Tow Rope, Fly-through, Walkthrough, Pilot-through, Move-the-World, Scale-the-World, Put-me-there, Put-me-there-WIM, Orbital Viewing, World-in-hand, Point-to-fly
- **Venue:** Home, Work, Museum, Arcade, Retail Store, Trade Show, Product Showroom, Classroom, School District, Outdoors, In-vehicle, Corporate Meeting Room, Physical Rehabilitation, Mental Therapist Office, Theme Park, Research Lab, Design Facility, Factory, Doctor's Office, Training Center

As you read through the applications in this book, take special mental note of how the developers chose to address such issues as input and navigation. Also keep in mind the constraints posed to the developers such as the application's display, venue, and goals. In developing your own applications, you may struggle with similar constraints. By looking at example applications and the lessons learned by other developers with similar constraints, you can optimize your users' experience.

We provide a complete system for exploring this taxonomy in the final chapter, but want to introduce the reader to this concept so that this framework can help you to interpret the information as it is presented.

ABOUT THE APPLICATIONS IN THIS BOOK

This book provides a glimpse into dozens of virtual reality applications. The applications are from a wide variety of sources, and cover a broad expanse of application areas. The applications in this book are referred to as *classic* applications in that they are some of the earlier examples of virtual reality put into use. It is important to note that each of the applications was chosen for a specific reason. While many modern-day applications utilize many of the same techniques as those covered in this book, we chose to go back to the beginning of the advent of virtual reality. It is in these early applications where one can discover not only good techniques and ideas, but more importantly the why of these ideas. Many aspects of these applications are seen in more modern applications, but it is often the case that the developers of the newer applications saw a technique in one of the more foundational applications and chose to implement it in their endeavor. Through personal experience, discussions with early developers, and review of extensive notes and documentation, our goal was to tease out the whys of the choices the developers made, and also to discover what was tried that didn't work as well as planned. In some cases, early technology was simply inadequate

to implement an idea, but in other cases it turned out that there were other reasons at play that drove the application designers to choose a different path. It is important to understand how the applications in this book were chosen for inclusion.

Each application in this book was included because it illustrated some particular item or idea. Although the applications covered in this book were primarily developed and deployed in the United States, we are in no way implying that the bulk of the work in early virtual reality applications was done in the United States alone. Rather, due to the realities of logistics and travel, we were better able to offer an in-depth look at those applications that were chosen according to the criteria they laid out when this book was conceived. There were a number of heuristics that we used in order to choose which applications would be discussed in this volume. The heuristics were as follows:

- *Heuristic 1: It must be possible to discuss the application in public.* We are aware of many other virtual reality applications that illustrate the same points as many of the applications described in this book but, due to the confidentiality of the information, could not be published.
- *Heuristic 2: If at all possible, we strove to include applications that we were able to experience themselves, and/or at least to have direct personal discussions with the application creators.* This heuristic unfortunately led to most of the applications covered to be U.S.-centric.
- *Heuristic 3: The applications should illustrate a wide variety of technological implementations.* That is, our goal was to include some applications that used head-based displays, and some that used projection displays. Some applications were included because they utilized some novel device or technique. Some applications used haptics or sounds, or other sensory feedback in a unique way. This heuristic helped to ensure that the coverage was not limited to CAVE™ applications, or head-mounted display applications, or any single technology that was prevalent at the time.
- *Heuristic 4: To the extent possible, include applications that were developed in different development environments, such as academia, government, and private sector laboratories.* This heuristic was to ensure a heterogeneous coverage of developers. Academics often attend different conferences, read different journals, and so on, than do industry developers. We believed that it was important to report on the work from all of those groups.
- *Heuristic 5: Include some applications that have ceased to exist, and some applications that have continued to be used, developed, commercialized, or otherwise exist today.* The important issue with this heuristic is the reason *why* some applications survive and some don't. In some cases an application has ceased to exist because it served its purpose, or a new solution to the problem it was addressing arose. In other cases, it is sometimes that the application was implemented in a way that caused the intended audience to not embrace the application. Likewise, some applications have survived because they genuinely aided in the solution of a difficult problem, there was commercial interest in the application, or the application developers saw the potential to fully realize the implementation.

The applications in this book have been organized into eight chapters. It is a difficult task to categorize the applications because any given application might be appropriate in more than one chapter. It is our hope that by presenting the highlights and problems with these applications that the reader can jump-start their own applications, get new ideas for techniques, and gain an overall appreciation for the breadth and depth of VR applications that came before.

If I have seen a little further it is by standing on the shoulders of giants.

– Isaac Newton

Throughout this book, we follow a standard template for describing each application.

Applications are divided into major applications and minor applications. For each major application, the following sections are included: Executive Summary, Introduction, Application Description, Representation of the Virtual World, Interaction with the Virtual World, Venue, VR System, Application Implementation, and Conclusion. Minor applications comprise a subset of these, including: Executive Summary, Application Description, VR System, and Conclusion.